

- ence on Learning Representations, 2025. URL <https://openreview.net/forum?id=I4e82CIDxv>.
- Meng, K., Bau, D., Andonian, A. J., and Belinkov, Y. Locating and editing factual associations in GPT. In Oh, A. H., Agarwal, A., Belgrave, D., and Cho, K. (eds.), *Advances in Neural Information Processing Systems*, 2022. URL <https://openreview.net/forum?id=-h6WAS6eE4>.
- Merullo, J., Eickhoff, C., and Pavlick, E. Circuit component reuse across tasks in transformer language models. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024. URL <https://openreview.net/forum?id=fpoAYV6Wsk>.
- Miller, J., Chughtai, B., and Saunders, W. Transformer circuit evaluation metrics are not robust. In *First Conference on Language Modeling*, 2024. URL <https://openreview.net/forum?id=zSf8PJyQb2>.
- Mills, E., Su, S., Russell, S., and Emmons, S. Almanacs: A simulatability benchmark for language model explainability. *CoRR*, abs/2312.12747, 2023.
- Mueller, A. Missed causes and ambiguous effects: Counterfactuals pose challenges for interpreting neural networks. In *ICML 2024 Workshop on Mechanistic Interpretability*, 2024. URL <https://openreview.net/forum?id=pJs3ZiKBM5>.
- Mueller, A., Brinkmann, J., Li, M., Marks, S., Pal, K., Prakash, N., Rager, C., Sankaranarayanan, A., Sharma, A. S., Sun, J., Todd, E., Bau, D., and Belinkov, Y. The quest for the right mediator: A history, survey, and theoretical grounding of causal interpretability. *CoRR*, abs/2408.01416, 2024. URL <https://arxiv.org/abs/2408.01416>.
- Nanda, N. Attribution Patching: Activation Patching At Industrial Scale, 2023. URL <https://www.neelnanda.io/mechanistic-interpretability/attribution-patching>.
- Nanda, N., Chan, L., Lieberum, T., Smith, J., and Steinhart, J. Progress measures for grokking via mechanistic interpretability. In *The Eleventh International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=9XFSbDPmdW>.
- Nikankin, Y., Reusch, A., Mueller, A., and Belinkov, Y. Arithmetic without algorithms: Language models solve math with a bag of heuristics. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=09Ytt26r2P>.
- Norlund, T., Hagström, L., and Johansson, R. Transferring knowledge from vision to language: How to achieve it and how to measure it? In Bastings, J., Belinkov, Y., Dupoux, E., Giulianelli, M., Hupkes, D., Pinter, Y., and Sajjad, H. (eds.), *Proceedings of the Fourth BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, pp. 149–162, Punta Cana, Dominican Republic, November 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.blackboxnlp-1.10. URL <https://aclanthology.org/2021.blackboxnlp-1.10/>.
- Olah, C., Cammarata, N., Schubert, L., Goh, G., Petrov, M., and Carter, S. Zoom in: An introduction to circuits. *Distill*, 2020. doi: 10.23915/distill.00024.001. <https://distill.pub/2020/circuits/zoom-in>.
- OpenAI. ChatGPT. <https://openai.com/chatgpt>, 2022.
- Paik, C., Aroca-Ouellette, S., Roncone, A., and Kann, K. The World of an Octopus: How Reporting Bias Influences a Language Model’s Perception of Color. In Moens, M.-F., Huang, X., Specia, L., and Yih, S. W.-t. (eds.), *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pp. 823–835, Online and Punta Cana, Dominican Republic, November 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.emnlp-main.63. URL <https://aclanthology.org/2021.emnlp-main.63/>.
- Pearl, J. Direct and indirect effects. In *Proceedings of the Seventeenth Conference on Uncertainty in Artificial Intelligence, UAI’01*, pp. 411–420, San Francisco, CA, USA, 2001. Morgan Kaufmann Publishers Inc. ISBN 1558608001.
- Prakash, N., Shaham, T. R., Haklay, T., Belinkov, Y., and Bau, D. Fine-tuning enhances existing mechanisms: A case study on entity tracking. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024. URL <https://openreview.net/forum?id=8sKcAWOf2D>.
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., Sutskever, I., et al. Language models are unsupervised multitask learners. Blog post, 2019. URL https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf.
- Räuber, T., Ho, A., Casper, S., and Hadfield-Menell, D. Toward transparent AI: A survey on interpreting the inner structures of deep neural networks. In *2023 IEEE Conference on Secure and Trustworthy Machine Learning, SaTML 2023, Raleigh, NC, USA, February 8-10, 2023*, pp. 464–483. IEEE, 2023. doi: 10.1109/SaTML54575.2023.

00039. URL <https://doi.org/10.1109/SaTML54575.2023.00039>.
- Riviere, M., Pathak, S., Sessa, P. G., Hardin, C., Bhupatiraju, S., Hussenot, L., Mesnard, T., Shahriari, B., Ramé, A., et al. Gemma 2: Improving open language models at a practical size. arXiv:2408.00118, 2024. URL <https://arxiv.org/abs/2408.00118>.
- Saphra, N. and Wiegrefe, S. Mechanistic? In *The 7th BlackboxNLP Workshop*, 2024. URL <https://openreview.net/forum?id=schAf4BPTd>.
- Schwettmann, S., Shaham, T. R., Materzynska, J., Chowdhury, N., Li, S., Andreas, J., Bau, D., and Torralba, A. FIND: A function description benchmark for evaluating interpretability methods. In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine, S. (eds.), *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*, 2023. URL http://papers.nips.cc/paper_files/paper/2023/hash/ef0164c1112f56246224af540857348f-Abstract-Datasets_and_Benchmarks.html.
- Sharkey, L., Chughtai, B., Batson, J., Lindsey, J., Wu, J., Bushnaq, L., Goldowsky-Dill, N., Heimersheim, S., Ortega, A., Bloom, J., Biderman, S., Garriga-Alonso, A., Conmy, A., Nanda, N., Rumbelow, J., Wattenberg, M., Schoots, N., Miller, J., Michaud, E. J., Casper, S., Tegmark, M., Saunders, W., Bau, D., Todd, E., Geiger, A., Geva, M., Hoogland, J., Murfet, D., and McGrath, T. Open problems in mechanistic interpretability. *CoRR*, abs/2501.16496, 2025. URL <https://arxiv.org/abs/2501.16496>.
- Shi, C., Beltran-Velez, N., Nazaret, A., Zheng, C., Garriga-Alonso, A., Jesson, A., Makar, M., and Blei, D. Hypothesis testing the circuit hypothesis in LLMs. In *ICML 2024 Workshop on Mechanistic Interpretability*, 2024. URL <https://openreview.net/forum?id=ibSNv9cldu>.
- Smolensky, P. Neural and conceptual interpretation of PDP models. In McClelland, J. L., Rumelhart, D. E., and the PDP Research Group (eds.), *Parallel Distributed Processing: Explorations in the Microstructure of Cognition: Psychological and Biological Models*, volume 2, pp. 390–431. MIT Press, 1986.
- Stolfo, A., Belinkov, Y., and Sachan, M. A mechanistic interpretation of arithmetic reasoning in language models using causal mediation analysis. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 7035–7052, 2023.
- Sundararajan, M., Taly, A., and Yan, Q. Axiomatic attribution for deep networks. In *Proceedings of the 34th International Conference on Machine Learning - Volume 70*, ICML’17, pp. 3319–3328. JMLR.org, 2017.
- Syed, A., Rager, C., and Conmy, A. Attribution patching outperforms automated circuit discovery. In Belinkov, Y., Kim, N., Jumelet, J., Mohebbi, H., Mueller, A., and Chen, H. (eds.), *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pp. 407–416, Miami, Florida, US, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.blackboxnlp-1.25. URL <https://aclanthology.org/2024.blackboxnlp-1.25/>.
- Tigges, C., Hollinsworth, O. J., Geiger, A., and Nanda, N. Linear representations of sentiment in large language models. *CoRR*, abs/2310.15154, 2023. URL <https://arxiv.org/abs/2310.15154>.
- Vig, J., Gehrmann, S., Belinkov, Y., Qian, S., Nevo, D., Singer, Y., and Shieber, S. Investigating gender bias in language models using causal mediation analysis. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H. (eds.), *Advances in Neural Information Processing Systems*, volume 33, pp. 12388–12401. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/file/92650b2e92217715fe312e6fa7b90d82-Paper.pdf.
- Wang, K. R., Variengien, A., Conmy, A., Shlegeris, B., and Steinhardt, J. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023. URL <https://openreview.net/forum?id=NpsVSN6o4ul>.
- Wiegrefe, S., Tafford, O., Belinkov, Y., Hajishirzi, H., and Sabharwal, A. Answer, assemble, ace: Understanding how LMs answer multiple choice questions. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=6NNA0MxhCH>.
- Wu, Z., Geiger, A., Icard, T., Potts, C., and Goodman, N. D. Interpretability at scale: Identifying causal mechanisms in alpaca. In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine, S. (eds.), *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*, 2023. URL http://papers.nips.cc/paper_files/paper/2023/hash/f6a8b109d4d4fd64c75e94aaf85d9697-Abstract-Conference.html.

- Wu, Z., Geiger, A., Arora, A., Huang, J., Wang, Z., Goodman, N., Manning, C. D., and Potts, C. pyvene: A library for understanding and improving pytorch models via interventions. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 3: System Demonstrations)*, pp. 158–165, 2024. URL <https://aclanthology.org/2024.naacl-demo.16/>.
- Wu, Z., Arora, A., Geiger, A., Wang, Z., Huang, J., Jurafsky, D., Manning, C. D., and Potts, C. AxBench: Steering LLMs? even simple baselines outperform sparse autoencoders. *CoRR*, abs/2501.17148, 2025. URL <https://arxiv.org/abs/2501.17148>.
- Yang, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Li, C., Liu, D., Huang, F., Wei, H., et al. Qwen2.5 technical report. arXiv:2412.15115, 2024. URL <https://arxiv.org/abs/2412.15115>.
- Zhang, F. and Nanda, N. Towards best practices of activation patching in language models: Metrics and methods. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=Hf17y6u9BC>.
- Zhang, W., Wan, C., Zhang, Y., ming Cheung, Y., Tian, X., Shen, X., and Ye, J. Interpreting and improving large language models in arithmetic calculation. In *Forty-first International Conference on Machine Learning*, 2024. URL <https://openreview.net/forum?id=Cf0tiepP8s>.
- Zhong, Z., Wu, Z., Manning, C., Potts, C., and Chen, D. MQuAKE: Assessing knowledge editing in language models via multi-hop questions. In Bouamor, H., Pino, J., and Bali, K. (eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 15686–15702, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.971. URL <https://aclanthology.org/2023.emnlp-main.971/>.